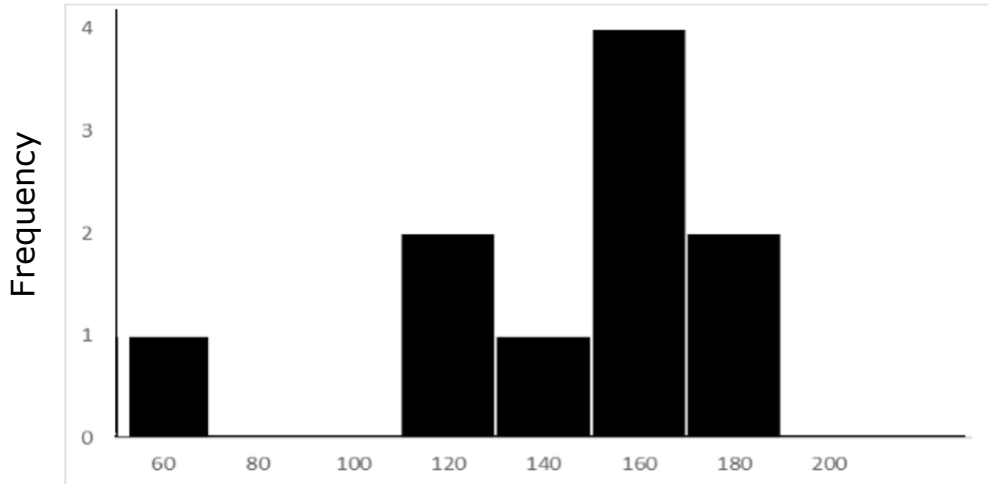


A realtor was asked to find the average price of homes for sale in their city, in thousands. The data the realtor found is shown below.



1. Using the histogram, is it possible for the value of the mean be determined? Explain.

No because the exact values are not given, just ranges.

2. Which data display(s) could be created with the data to find the value of the mean?

Dot plot or steam-and-leaf plot

3. Estimate the median of the data.

\$160,000

4. Are there any outliers in the data set?

Yes, \$60,000

5. Which measure of variation would you use to summarize the data?

With this graph, IQR would be best if the 1st and 3rd quartile values could be determined.

Kaiden asked his classmates what their favorite numbers were. He recorded their answers in the stem-leaf-plot shown below.

Stem	Leaf
1	8, 9
2	5, 8
3	0, 0, 1, 2, 2, 6, 6, 6, 7, 7, 7, 8, 9
4	0, 1, 2, 8
5	1, 2, 5, 8, 8, 9, 9
6	0, 4
Key: 1 8 = 18	

6. Kaiden said the average favorite number recorded was 42.3. Explain whether you agree or disagree with Kaiden?

$\text{mean} = \frac{1228}{30} \approx 40.9$ I disagree because the average, or mean, of the data is approximately 40.9.

7. Find the range of the data.

$$64 - 18 = 46$$

8. Find the IQR of the data.

$$52 - 32 = 20$$

9. Which measure of variation best describes the data? Explain your reasoning.

IQR because data is slightly skewed.

10. Describe what effect outliers have on the different measures of variation.

Outliers can skew the range, making it show a larger variation.
Outliers affect the IQR less which is why it is the preferable measure of variation when outliers are present.